

Review retrospective:
*the multi-index affine-bilinear covariance — and a docstring
that called itself 2D*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

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Abstract

A soundness review of `Laplace/Multi/AffineBilinearCov.lean`: the multi-index Gibbs covariance is affine-bilinear, $\text{Cov}_t[a_1\varphi + b_1, a_2\psi + b_2] = a_1a_2 \text{Cov}_t[\varphi, \psi]$ on $w \in (\iota \rightarrow \mathbb{R})$. Result: **a clean fidelity pass with two landed edits** — a genuinely dead `open Real` and a docstring accuracy fix (the file is polymorphic in `[Fintype ι]` but called itself “the 2D mirror”). This is the first review into the `laplace Multi/` multivariate layer beyond the previously-audited `GaussianIBP`.

1 Setting

The primer’s Stage 3 is the multivariate covariance expansion. Its algebraic base is the multi-index `gibbsCov` bilinearity, assembled in `Multi/Basic.lean` (additivity, scalar-linearity, constant-vanishing, symmetry, each side). This file is its *first consumer*: the affine-input reduction, the multi-index mirror of the one-dimensional “Tide I3” affine-bilinear result. Reviewing it checks that the consumer composes the base algebra correctly, and incidentally exercises the `Multi/` layer for the first time since the `GaussianIBP` review.

2 What was reviewed

Three public theorems: `gibbsCov_affine_left` ($\text{Cov}[a\varphi + b, \psi] = a \text{Cov}$), `gibbsCov_affine_right` (the right-slot mirror via `gibbsCov_symm`), and the headline `gibbsCov_affine_affine` ($a_1a_2 \text{Cov}$). Checked against the module docstring, the `Multi/Basic` algebra lemmas the proofs chain, the 1D sibling `Laplace/AffineBilinearCov.lean`, and the tide retrospective.

3 Fidelity / soundness findings

Sound; GPT found no mismatch. The right-hand sides are exactly a , a , and a_1a_2 times the base covariance — one factor of each linear scalar, with *no* surviving b_1/b_2 constant terms (the constants drop because the covariance of anything with a constant is zero). The four integrability hypotheses ($h_{\text{exp}}, h_\varphi, h_\psi, h_{\varphi\psi}$) are exactly the base interface the chained algebra requires: none too few (the proofs derive the affine-transformed integrabilities internally, by `const_mul/add`), and none redundant in the public signature — all four are genuinely used. GPT’s one proof-side note (that

`affine_right` invokes `gibbsCov_symm` without separately deriving an affine-slot integrability) is consistent with how `symm` is stated and is explicitly *not* a statement-level mismatch.

4 Simplifications applied

Two landed.

- **Dead open Real.** Every exponential is written fully qualified as `Real.exp`; the only bare “exp” is in a docstring comment, and no other `Real`-namespace name or notation appears. `open MeasureTheory` stays — `Integrable` is used unqualified throughout the statements. Removed and rebuild-arbitrated.
- **Docstring accuracy: “2D mirror” → “multi-index mirror”.** The file is polymorphic in `[Fintype ι]` — it covers every finite dimension, not just $\iota = \text{Fin } 2$ — and already calls itself “the multi-index analogue” in its opening sentence. The trailing “the 2D mirror of I3” was an internal inconsistency, narrower than what is proved; corrected to match.

Both are comment/open-line only: zero proof-body, zero signature change. Baseline green (2544 jobs) → both edits → rebuild green (2544), all three signatures byte-identical, GPT-affirmed, fast-forward-merged. GPT’s other four proposals — eliminate the `rfl eq` boilerplate, `simpa-collapse` the two `funext;ring` integrability blocks, compress `affine_right`’s rewrite chain, and rename the local hypotheses — are tactic-golf or cosmetic and were declined.

5 Disagreements and open questions

None on the mathematics.

6 Lessons

- **A docstring that contradicts itself is a reliable place to find an accuracy fix.** The opening line (“multi-index analogue”) and the closing line (“2D mirror”) disagreed about the file’s own generality; the closing one was wrong. When two parts of one docstring describe the same object differently, at least one is stale — check which against the actual type signature (`[Fintype ι]` here settles it).
- **The dead-open grep is now a fast, two-confirmation routine.** Across the recent laplace reviews the pattern is settled: grep the file for any bare name/notation the open provides (`Integrable` for `MeasureTheory`, bare `exp/π` for `Real`), and if the static check and the independent reviewer agree the open is unused, the rebuild confirms it definitively. Here `Real` was dead and `MeasureTheory` load-bearing, decided in one pass.

7 Follow-ups

- The sibling `Multi/QuadraticApprox.lean` (the other small `Multi/` foundation file) is the natural next review; the large `Multi/Covariance*.lean` files (3998–19611 lines) each exceed a single review-tide scope and would need a sectioned approach.